Amazon Indian Sales Analysis

1. Project Overview

E-commerce managers often struggle to connect high-level financial performance with product-level demand and order logistics. This project addresses that gap by consolidating revenue, geographic sales, product categories, and fulfillment performance into an interactive, 3-page executive Power BI dashboard.

The resulting solution enables business directors, sales managers, and fulfillment leads to track overall profitability, evaluate category performance, analyze regional demand across India, and optimize order delivery workflows from a single unified view.

2. Dataset Summary

- Rows: 50,000
- Columns: 15
  - Key Features:

- Order Details**:** order_id, order_date, units_sold
- Product & Category: category, brand, price_category
- Financial Metrics: total_revenue, net_revenue, aov, return_rate
- Fulfillment & Payment: delivery_status, payment_method
- Geographic Data: State / Region breakdown (India)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the 50,000 Amazon Indian sales and product records into a Data Frame using `pandas`

```
df = pd.read_csv("amazon_india_raw_products.csv")

print(df.head(30).to_string())
```

	order_id	order_date	customer_id	customer_state	product_id	product_name	category	brand	price	quantity	discount_percent	payment_method	rating	review_count	delivery_status
0	ORD-133553	2025-11-30	CUST-21239	NaN	PROD-10476	LG Odio Numquam	Clothing & Apparel	LG	40398.62	4	15	Net Banking	3.7	2744	returned
1	ORD-109427	06-03-2025	CUST-26699	Uttar Pradesh	PROD-10464	Apple Cum At	Electronics	apple	21706.45	1	20	UPI	1.4	583	Returned
2	ORD-100199	2026-02-20	CUST-25088	Tamil Nadu	PROD-10940	Philips Ducimus Error	Clothing & Apparel	philips	41965.0	1	30	UPI	2.5	3261	Delivered
3	ORD-112447	30-Dec-2024	CUST-21637	Telangana	PROD-10414	Apple Aliquam Possimus	Electronics	Apple	11892.04	4	5	CC	2.3	1702	Pending
4	ORD-139489	2025-02-20	CUST-24837	telangana	PROD-10157	Xiaomi Esse Aspernatur	Electronics	Xiaomi	36042.76	2	15	Net Banking	2.8	1254	Pending
5	ORD-142724	2026-08-23	CUST-23009	West Bengal	PROD-10055	Puma Error Ullam	Home & Kitchen	Puma	12123.73	2	5	UPI	4.7	1989	Cancelled
6	ORD-110822	2025-04-08	CUST-27795	NaN	PROD-10825	LG Quam Cum	Electronics	lg	2454.43	1	20	NaN	5.0	3425	cancelled
7	ORD-123339	2025-03-08	CUST-21796	delhi	PROD-10410	Nike Odio Exercitationem	Sports & Fitness	Nike	3347.21	5	50	UPI	4.5	268	Delivered
8	ORD-104144	2026-02-13	CUST-23082	Uttar Pradesh	PROD-10808	LG Dicta Tempora	Home & Kitchen	lg	20908.11	1	0	C.O.D.	3.9	2373	Cancelled
9	ORD-136958	2026-03-01	CUST-24335	DELHI	PROD-10834	Puma Ab Enim	ELECTRONICS	PUMA	INR 9086.99	3	10	DEBIT CARD	3.6	2500	Shipped
10	ORD-143106	09-Mar-2026	CUST-21128	West Bengal	PROD-10087	Sony Dolor Minima	CLOTHING & APPAREL	SONY	43258.59	2	30	upi	1.6	1533	Cancelled
11	ORD-138695	2025-11-30	CUST-27174	Gujarat	PROD-10841	Realme Modi Quibusdam	Electronics	realme	1803.42	3	30	Credit Card	1.4	1567	Cancelled
12	ORD-106188	13/06/2025	CUST-22177	Tamil Nadu	PROD-10920	Xiaomi Molestiae Id	electronics	Xiaomi	4855.33	3	10	UPI	NaN	790	returned
13	ORD-101414	2026-07-25	CUST-22058	NaN	PROD-10076	Philips Vero Quidem	Beauty & Personal Care	Philips	23456.83	2	30	Net Banking	NaN	273	Delivered
14	ORD-118471	2025-08-27	CUST-25215	Karnataka	PROD-10507	LG Omnis Voluptas	Beauty & Personal Care	LG	28754.3	2	0	Credit Card	2.9	2809	Returned
15	ORD-129282	04-06-2026	CUST-22532	Maharashtra	PROD-10272	Sony Quod Debitis	Electronics	NaN	43660.44	2	30	cash on delivery	3.6	2723	Cancelled
16	ORD-115177	2026-01-29	CUST-21617	Gujarat	PROD-10196	boAt In Ut	SPORTS & FITNESS	boAt	10853.85	1	10	Net Banking	1.9	1937	Returned
17	ORD-134304	27-Mar-2026	CUST-20098	Telangana	PROD-10537	LG Explicabo Iste	Beauty & Personal Care	LG	25167.09	5	0	Credit Card	2.5	2055	Cancelled
18	ORD-112609	30/11/2025	CUST-24385	Telangana	PROD-10959	Philips Quibusdam Excepturi	Beauty & Personal Care	Philips	39001.62	1	0	cash on delivery	4.5	388	Cancelled
19	ORD-112144	NaN	CUST-23282	Uttar Pradesh	PROD-10147	Adidas Placeat Perspiciatis	Electronics	Adidas	5597.45	4	10	Credit Card	2.9	3102	Shipped
20	ORD-106113	2024-11-07	CUST-23246	Uttar Pradesh	PROD-10901	Samsung Sed Pariatur	Beauty & Personal Care	samsung	7046.08	2	10	Net Banking	1.2	914	pending
21	ORD-115908	2025-09-17	CUST-23779	uttar pradesh	PROD-10321	Prestige Placeat Sunt	Home & Kitchen	Prestige	31206.17	1	50	Credit Card	1.4	991	Delivered
22	ORD-100821	25-Oct-2025	CUST-27813	Telangana	PROD-10839	Philips Praesentium Non	clothing & apparel	Philips	12525.1	-1	50	CASH ON DELIVERY	2.7	340	CANCELLED
23	ORD-115118	09/01/2026	CUST-27382	Delhi	PROD-10788	Nike Quae Error	electronics	Nike	16667.07	1	30	UPI	1.0	1907	Cancelled
...															
26	ORD-142111	2025-01-12	CUST-23034	delhi	PROD-10755	Nike Harum Soluta	Clothing & Apparel	Nike	14093.52	1	15	NaN	3.6	3118	Returned
27	ORD-130188	2025-12-28	CUST-22233	Telangana	PROD-10925	LG Dicta Laudantium	Electronics	LG	44759.6	2	10	Credit Card	3.5	3269	Delivered
28	ORD-137237	15-Jan-2026	CUST-27100	Tamil Nadu	PROD-10574	LG Unde Porro	Beauty & Personal Care	LG	₹23190.66	1	20	Debit Card	2.4	653	returned
29	ORD-133109	12-Dec-2024	CUST-23239	MAHARASHTRA	PROD-10763	boAt Cum Ullam	HOME & KITCHEN	NaN	2293.91	1	5	Net Banking	4.4	356	Shipped

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

• **Initial Exploration:** Used **df.head()** to get a basic overview about the Columns and its Rows

```
df.info()
```

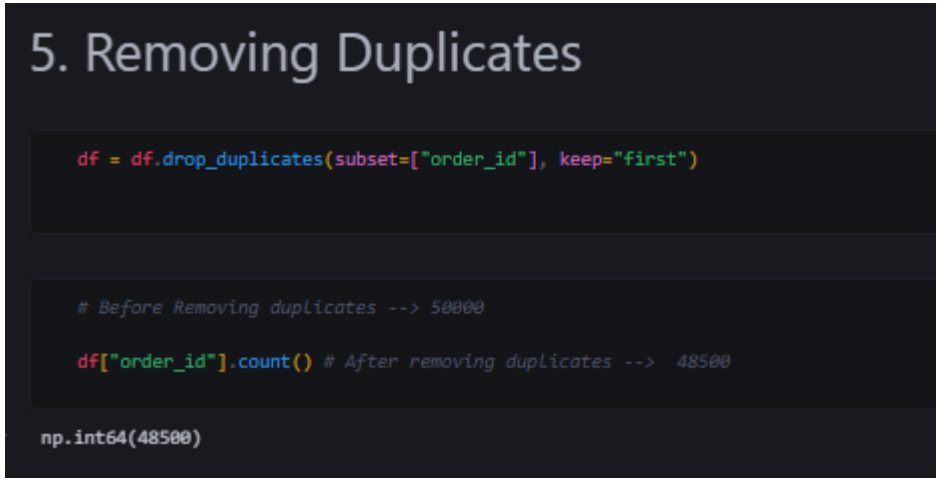
```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   order_id          50000 non-null  str
 1   order_date        48523 non-null  str
 2   customer_id       50000 non-null  str
 3   customer_state    46927 non-null  str
 4   product_id        50000 non-null  str
 5   product_name      50000 non-null  str
 6   category          50000 non-null  str
 7   brand             48006 non-null  str
 8   price             47567 non-null  str
 9   quantity          50000 non-null  int64
 10  discount_percent  50000 non-null  int64
 11  payment_method    48550 non-null  str
 12  rating            45969 non-null  str
 13  review_count      50000 non-null  int64
 14  delivery_status   48987 non-null  str
dtypes: int64(3), str(12)
memory usage: 5.7 MB
```

• **Checking for Null & Duplicate Value:** Used **df.apply(lambda col : col.duplicated().sum())** to check the total number of duplicate row in the Data set

```
    # Returns a Series with duplicate count for each column
    df.apply(lambda col: col.duplicated().sum())
```

```
order_id             1500
order_date          47081
customer_id         42024
customer_state      49951
product_id          49000
product_name        49002
category            49964
brand               49929
price               47130
quantity            49992
discount_percent    49993
payment_method      49964
rating              49957
review_count        46499
delivery_status     49969
dtype: int64
```

• **Data Cleaning:** Handled missing values, Purged duplicates (`drop_duplicates()`)



• **Standardization:**

6. Standardization

```
print(df.head(10).to_string())
```

```
   order_id  order_date customer_id customer_state product_id            product_name          category     brand       price  quantity  discount_percent payment_method rating  review_count delivery_status
0  ORD-133553  2025-11-30  CUST-21239            NaN  PROD-10476         LG Odio Numquam  Clothing & Apparel        LG     40398.62         4                15    Net Banking     3.7          2744        returned
1  ORD-109427  06-03-2025  CUST-26699     Uttar Pradesh  PROD-10464            Apple Cum At       Electronics     apple     21706.45         1                20            UPI     1.4           583        Returned
2  ORD-100199  2026-02-20  CUST-25088        Tamil Nadu  PROD-10940     Philips Ducimus Error  Clothing & Apparel   philips      41965.0         1                30            UPI     2.5          3261       Delivered
3  ORD-112447  30-Dec-2024  CUST-21637         Telangana  PROD-10414    Apple Aliquam Possimus       Electronics     Apple     11892.04         4                 5             CC     2.3          1702         Pending
4  ORD-139489  2025-02-20  CUST-24837         telangana  PROD-10157    Xiaomi Esse Aspernatur       Electronics    Xiaomi     36042.76         2                15    Net Banking     2.8          1254         Pending
5  ORD-142724  2026-08-23  CUST-23009       West Bengal  PROD-10055         Puma Error Ullam     Home & Kitchen      Puma     12123.73         2                 5            UPI     4.7          1989       Cancelled
6  ORD-110822  2025-04-08  CUST-27795            NaN  PROD-10825             LG Quam Cum       Electronics        lg      2454.43         1                20            NaN     5.0          3425       cancelled
7  ORD-123339  2025-03-08  CUST-21796          delhi  PROD-10410  Nike Odio Exercitationem   Sports & Fitness      Nike      3347.21         5                50            UPI     4.5           268       Delivered
8  ORD-104144  2026-02-13  CUST-23082    Uttar Pradesh  PROD-10808         LG Dicta Tempora     Home & Kitchen        lg     20908.11         1                 0           C.O.D.     3.9          2373       Cancelled
9  ORD-136958  2026-03-01  CUST-24335           DELHI  PROD-10834             Puma Ab Enim         ELECTRONICS      PUMA  INR 9086.99         3                10     DEBIT CARD     3.6          2500         Shipped
```

```
# Standardize the Column

# "customer_state", "delivery_status"

df["customer_state"] = df["customer_state"].str.strip().str.title()
df["delivery_status"] = df["delivery_status"].str.strip().str.title()



print(df.head(10).to_string())
```

```
   order_id  order_date customer_id customer_state  product_id            product_name          category     brand       price  quantity  discount_percent payment_method rating  review_count delivery_status
0  ORD-133553  2025-11-30  CUST-21239            NaN  PROD-10476         LG Odio Numquam  Clothing & Apparel        LG     40398.62         4                15    Net Banking     3.7          2744        Returned
1  ORD-109427  06-03-2025  CUST-26699  Uttar Pradesh  PROD-10464            Apple Cum At       Electronics     apple     21706.45         1                20            UPI     1.4           583        Returned
2  ORD-100199  2026-02-20  CUST-25088     Tamil Nadu  PROD-10940     Philips Ducimus Error  Clothing & Apparel   philips      41965.0         1                30            UPI     2.5          3261       Delivered
3  ORD-112447  30-Dec-2024  CUST-21637      Telangana  PROD-10414    Apple Aliquam Possimus       Electronics     Apple     11892.04         4                 5             CC     2.3          1702         Pending
4  ORD-139489  2025-02-20  CUST-24837      Telangana  PROD-10157    Xiaomi Esse Aspernatur       Electronics    Xiaomi     36042.76         2                15    Net Banking     2.8          1254         Pending
5  ORD-142724  2026-08-23  CUST-23009    West Bengal  PROD-10055         Puma Error Ullam     Home & Kitchen      Puma     12123.73         2                 5            UPI     4.7          1989       Cancelled
6  ORD-110822  2025-04-08  CUST-27795            NaN  PROD-10825             LG Quam Cum       Electronics        lg      2454.43         1                20            NaN     5.0          3425       Cancelled
7  ORD-123339  2025-03-08  CUST-21796         Delhi  PROD-10410  Nike Odio Exercitationem   Sports & Fitness      Nike      3347.21         5                50            UPI     4.5           268       Delivered
8  ORD-104144  2026-02-13  CUST-23082  Uttar Pradesh  PROD-10808         LG Dicta Tempora     Home & Kitchen        lg     20908.11         1                 0           C.O.D.     3.9          2373       Cancelled
9  ORD-136958  2026-03-01  CUST-24335         Delhi  PROD-10834             Puma Ab Enim         ELECTRONICS      PUMA  INR 9086.99         3                10     DEBIT CARD     3.6          2500         Shipped
```

- **Handling NaN:**

```
missing = df.isna().sum()

missing
```

```
order_id              0
order_date         1430
customer_id           0
customer_state     2989
product_id            0
product_name          0
category              0
brand              1928
price              2360
quantity              0
discount_percent      0
payment_method     1406
rating             3911
review_count          0
delivery_status     979
dtype: int64
```

```
# Handling the Critical Columns first

df = df.dropna(subset=["order_date", "price"])
```

```
missing = df.isna().sum()

missing
```

```
order_id              0
order_date            0
customer_id           0
customer_state     2755
product_id            0
product_name          0
category              0
brand              1780
price                 0
quantity              0
discount_percent      0
payment_method     1307
rating             3603
review_count          0
delivery_status     909
dtype: int64
```

```
# Filling the nan in textual column with "unknown"

text_column = ["customer_state", "brand", "payment_method", "delivery_status"]
df[text_column] = df[text_column].fillna("unknown")
```

```
missing = df.isna().sum()

missing
```

```
order_id              0
order_date            0
customer_id           0
customer_state        0
product_id            0
product_name          0
category              0
brand                 0
price                 0
quantity              0
discount_percent      0
payment_method        0
rating             3603
review_count          0
delivery_status       0
dtype: int64
```

```python
# 1. Strip any leftover text (e.g., '4.5 Stars' -> '4.5') and convert to float
df["rating"] = pd.to_numeric(
    df["rating"].astype(str).str.replace(r"[^\d.]", "", regex=True),
    errors="coerce",
)

# 2. Calculate median (ignores existing NaNs automatically)
rating_median = df["rating"].median()

# 3. Fill missing values with the calculated median
df["rating"] = df["rating"].fillna(rating_median)
```

[25]

```python
missing = df.isna().sum()

missing
```

[26]

```
order_id            0
order_date          0
customer_id         0
customer_state      0
product_id          0
product_name        0
category            0
brand               0
price               0
quantity            0
discount_percent    0
payment_method      0
rating              0
review_count        0
delivery_status     0
dtype: int64
```

```
# 1. Strip all non-numeric characters except decimals
df["price"] = df["price"].astype(str).str.replace(r"[^\d.]", "", regex=True)

# 2. Convert to float cleanly
df["price"] = pd.to_numeric(df["price"], errors="coerce")

# Standardize all mixed date formats into standard datetime objects (YYYY-MM-DD)
df["order_date"] = pd.to_datetime(
    df["order_date"], format="mixed", dayfirst=True, errors="coerce"
)
```

```
# Check that data types changed from 'object' to 'datetime64[ns]' and 'float64'
print(df[["order_date", "price"]].dtypes)

# Preview the clean rows
df[["order_date", "price"]].head(10)
```

```
order_date    datetime64[us]
price                float64
dtype: object
```

- **Changing the Data Types:**

9. Changing the Data types

[31]
```python
# quantity --> int64 to

df = df.astype(
    {
        "quantity": "int64",
        "price": "float64",
        "customer_state": "string",
    }
)
```

[32]
```python
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 44775 entries, 0 to 49999
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   order_id          44775 non-null  str
 1   order_date        44775 non-null  datetime64[us]
 2   customer_id       44775 non-null  str
 3   customer_state    44775 non-null  string
 4   product_id        44775 non-null  str
 5   product_name      44775 non-null  str
 6   category          44775 non-null  str
 7   brand             44775 non-null  str
 8   price             44775 non-null  float64
 9   quantity          44775 non-null  int64
 10  discount_percent  44775 non-null  int64
 11  payment_method    44775 non-null  str
 12  rating            44775 non-null  float64
 13  review_count      44775 non-null  int64
 14  delivery_status   44775 non-null  str
dtypes: datetime64[us](1), float64(2), int64(3), str(8), string(1)
memory usage: 5.5 MB
```

• **Feature Engineering :**

10. Feature Engineering

```
print(df.head().to_string())
```

```
  order_id order_date customer_id customer_state  product_id           product_name           category    brand     price  quantity  discount_percent payment_method  rating  review_count delivery_status
0  ORD-133553 2025-11-30  CUST-21239        unknown  PROD-10476        LG Odio Numquam  Clothing & Apparel       LG  40398.62         4                15    Net Banking     3.7          2744         Returned
1  ORD-109427 2025-03-06  CUST-26699  Uttar Pradesh  PROD-10464           Apple Cum At        Electronics    apple  21706.45         1                20            UPI     1.4           583         Returned
2  ORD-100199 2026-02-20  CUST-25088     Tamil Nadu  PROD-10940   Philips Ducimus Error  Clothing & Apparel  philips  41965.00         1                30            UPI     2.5          3261        Delivered
3  ORD-112447 2024-12-30  CUST-21637      Telangana  PROD-10414  Apple Aliquam Possimus        Electronics    Apple  11892.04         4                 5             CC     2.3          1702          Pending
4  ORD-139489 2025-02-20  CUST-24837      Telangana  PROD-10157  Xiaomi Esse Aspernatur        Electronics   Xiaomi  36042.76         2                15    Net Banking     2.8          1254          Pending
```

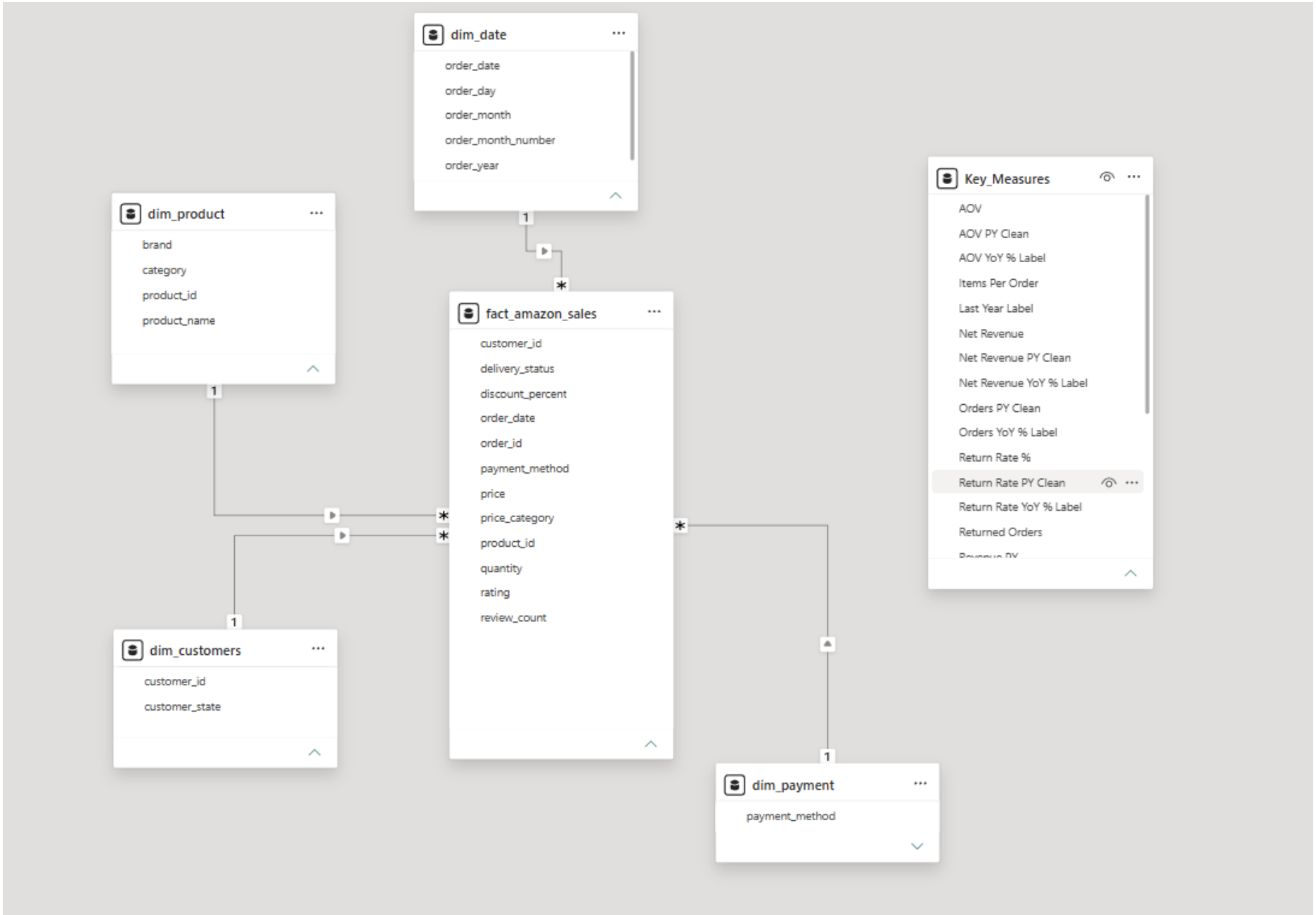
```
conditions = [

    (df["price"] < 1000),
    (df["price"] >= 1000) & (df["price"] < 5000),
    (df["price"] >= 5000) & (df["price"] < 20000),
    (df["price"] >= 20000),
]

choices = ["Budget", "Economy", "Mid-Range", "Premium"]

df["price_category"] = np.select(conditions, choices, default="unknown")
```

4. Power BI Data Model



Amazon Indian Sales Analysis

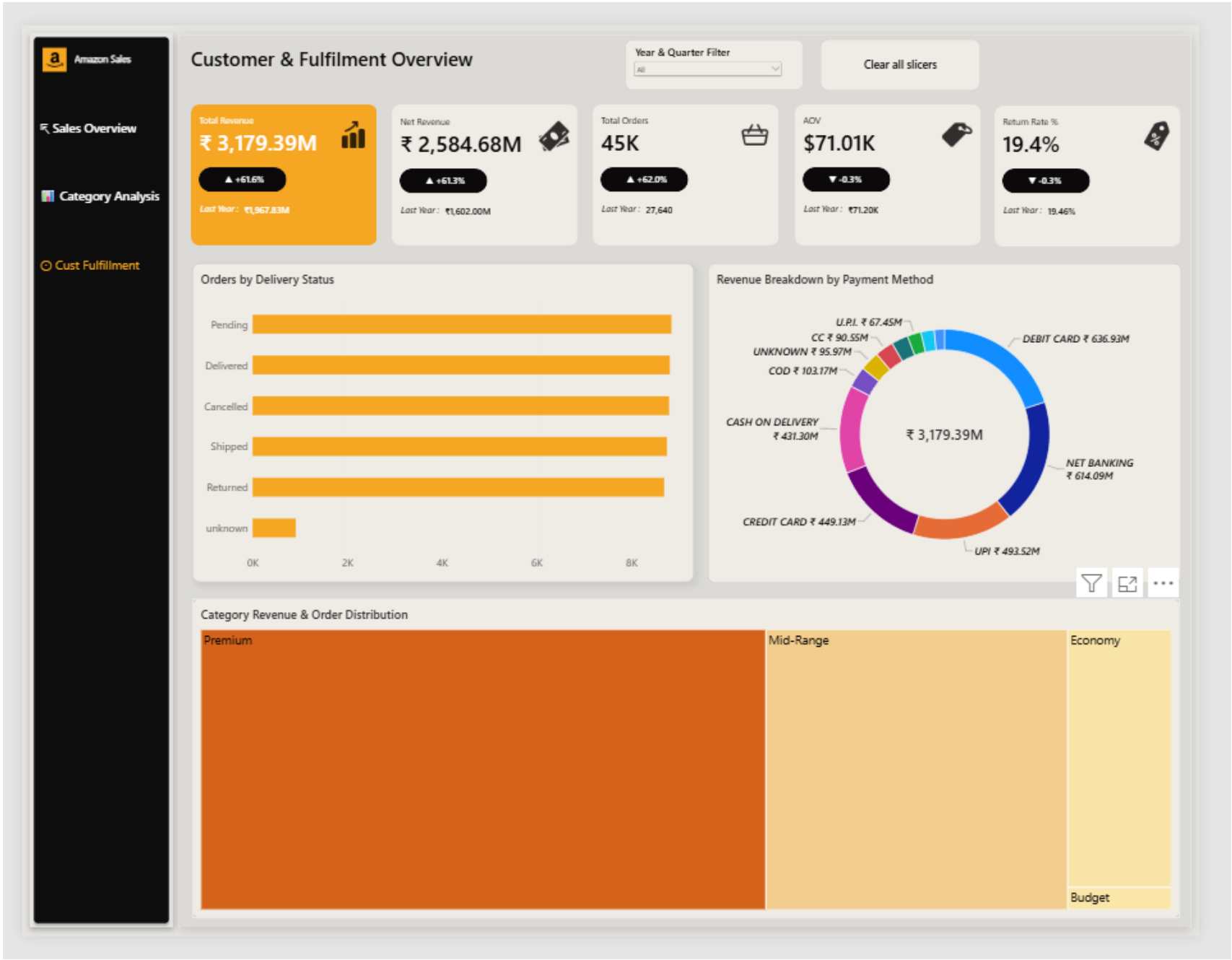
6. Dashboard



Amazon Indian Sales Analysis



Amazon Indian Sales Analysis



Amazon Indian Sales Analysis

7. Strategic Recommendation

Based on the multi-dimensional analysis of the 50,000 sales records, the following data-driven strategies are recommended for executive leadership, sales, and supply chain teams:

1. Supply Chain & Fulfillment Optimization • Mitigate High-Risk Return Zones: Implement targeted quality control checks and standardized size guides for categories with elevated return rates (such as apparel) to reduce reverse logistics costs. • Streamline Regional Warehousing: Capitalize on the high concentration of orders from metropolitan states by optimizing inventory placement in local fulfillment hubs, reducing delivery times for top-selling SKUs.
2. Payment & Cash Flow Management • Incentivize Digital Payments: Since digital channels (UPI and Net Banking) show healthier completion rates compared to Cash on Delivery (COD), introduce micro-incentives (e.g., instant wallet cashbacks) for prepaid orders to drive down cancellation rates. • Reduce COD Risk: Apply stricter order-verification protocols for high-value Cash on Delivery transactions in emerging tier-2 and tier-3 regions to minimize last-mile return overhead.
3. Commercial & Pricing Strategy • Capitalize on High-AOV Categories: Allocate higher marketing and inventory budgets toward top-performing categories (like Electronics) which yield superior Average Order Values and strong revenue contributions. • Refine Discount Thresholds: Restructure promotional campaigns so that high discounts (>20%) are paired with cross-selling bundles to protect profit margins while driving volume.